

DEPLOYERS: Comprehensive Analysis of a Darwinian Agent-Based Macroeconomic Simulator

Executive Summary

DEPLOYERS represents a novel approach to macroeconomic modeling through agent-based simulation that deploys an economic system from scratch to reproduce a target Social Accounting Matrix (SAM) of a real country. Unlike conventional economic models grounded in established theoretical frameworks, this simulator explores the emergent behavior of common-sense agents who operate without knowledge of economic theories, reflecting the decision-making patterns of actual households and small firm owners.

The system's distinctive characteristic lies in its **Darwinian evolutionary mechanism**: the economy does not begin with predefined firms but rather develops them endogenously through a process of firm creation, competition, and selective survival. An **assisted production stage** provides initial support to nascent firms, followed by a **calibration period** that guides the system toward a quasi-steady state with increasing GDP that matches empirical data from target economies.

PART I: SYSTEM ARCHITECTURE

1.1 Foundational Design Philosophy

The DEPLOYERS architecture embodies three core principles that distinguish it from traditional macroeconomic models:

Emergence Over Imposition: Rather than imposing equilibrium conditions or assuming rational expectations, the system allows economic structures to emerge from the bottom-up interactions of boundedly rational agents. Firms are not instantiated at initialization; they arise organically when agents accumulate sufficient wealth and perceive market opportunities.

SAM-Driven Calibration: The target SAM, typically derived from FIGARO inter-country input-output tables, serves as the empirical anchor. The system continuously adjusts parameters during the calibration phase to guide simulated aggregate flows toward these real-world values, ensuring that the emergent microeconomic behavior produces realistic macroeconomic outcomes.

Darwinian Selection: Firms face authentic survival pressures. Those unable to maintain positive wealth over consecutive periods are dismantled, with their assets distributed. This creates a natural selection mechanism where successful business models proliferate while unsustainable ones exit, mimicking real market dynamics without externally imposed closure rules.

1.2 Core Agent Hierarchy

The system implements a sophisticated class hierarchy rooted in a base CAgent class, with specialized behaviors for each economic role:

CAgent (Base Class): Provides fundamental capabilities shared across all economic actors, including cash management, goods inventory tracking, price memory for each good type, bank account operations, and transaction execution. The base class implements the core economic accounting logic that ensures conservation of value throughout the simulation.

CWorker (Households): Represents individual economic agents who supply labor, consume goods and services, accumulate wealth through savings and financial asset ownership, and entrepreneurially create new firms when conditions permit. Workers are initialized with zero wealth and gradually accumulate resources through employment and capital ownership. They maintain consumption preferences and adaptively adjust their price expectations based on market interactions.

CProducer (Firms): Autonomous production units that hire workers from the labor market, purchase intermediate inputs from other producers and external sectors, execute production using Leontief technology with both labor and capital constraints, set prices through adaptive markup rules over production costs, and manage working capital through commercial banking relationships. Producers track comprehensive operational metrics including production history, demand forecasts, inventory levels, and financial performance.

CGovernment: A singular institutional agent that collects taxes on production, products, and labor compensation, purchases goods and services according to the target SAM, provides unemployment transfers when activated, manages public debt through bond issuance to the central bank, and maintains fiscal accounts. The government's consumption patterns are calibrated during the adjustment period to match empirical government expenditure.

CExtSect (External Sectors):** *Specialized producers representing foreign trade partners, which supply imports to domestic producers and households with elastic supply curves and consume domestic exports with demand determined by the target SAM. Each external sector (which may represent individual countries or aggregated trading partners) maintains separate accounts for bilateral trade flows. These sectors operate without standard production constraints, essentially providing infinite supply at their price points.*

CBank and CCentralBank: Financial intermediaries that accept deposits from agents with positive cash balances, extend credit lines to firms needing working capital, calculate interest rates based on default risk assessment using equity ratios, and manage reserve requirements. The central bank also conducts open market operations with the government to support public debt management.

1.3 Architectural Components and Data Structures

Social Accounting Matrix (CSAM):** *This represents the heart of the system's structure. The SAM is constructed from input-output tables and defines the complete accounting framework. It specifies inter-sectoral flows through intermediate consumption matrices, value-added components by sector including compensation of employees and gross operating surplus, final demand by institutional sectors encompassing household consumption, government expenditure, and gross fixed capital formation, tax structures including production taxes and product taxes, and external sector transactions representing imports and exports by partner country.*

The SAM provides both the initial production coefficients and the target aggregate values that guide calibration. Each account in the SAM becomes either a producer type or an institutional account category in the simulation.

Product and Producer Specifications:** *For each production sector, the system maintains detailed technical coefficients derived from the SAM. The CProductSpecs class stores the intermediate input requirements per unit of output, labor requirements measured in compensation of employees per unit, capital requirements reflected in gross operating surplus per unit, and production tax rates by tax category. The CProducerSpecs class associates each producer type with its product specification and tracks initial gross output levels.*

These specifications enable the simulation of realistic production functions where output is constrained by the minimum of available inputs when evaluated in Leontief proportions, available labor adjusted for compensation levels, and existing fixed capital stock adjusted for capital productivity.

Economic Data Aggregation (CData):** *This component continuously accumulates simulation statistics required for analysis and calibration. During each simulation month, it tracks nominal GDP computed from final expenditure components, sectoral gross output and value added, unemployment rates and labor market statistics, price indices for products and consumer baskets, firm demographics including births, deaths, and size distributions, wealth distributions for both households and firms, and financial market indicators such as interest rates and credit volumes.*

The data aggregator implements moving averages for key variables to smooth short-term fluctuations and provide stable signals for adaptive control mechanisms. It also manages the construction of monthly SAM

matrices that record actual transaction flows, enabling comparison with target values.

Neighborhood Network (CNeighborsVectorManager):** *Rather than assuming perfect information or random matching, the system implements a spatial network structure. Each agent is assigned an identifier that determines its position in a conceptual economic space. Agents primarily interact with neighbors within their locality, creating realistic search frictions. Workers seek employment from nearby producers, and producers search for suppliers and customers within their neighborhoods. This network structure generates emergent spatial clustering of economic activity and allows for local market imbalances even as the aggregate system approaches steady state.*

Random Number Generation: Deterministic reproducibility is ensured through carefully managed random number generation. The system uses the Mersenne Twister algorithm with explicit state management. All stochastic elements including agent initialization sequence, transaction ordering, firm creation timing, and behavioral noise are controlled by this generator. The random state can be saved and restored, enabling exact replication of simulation runs for debugging and analysis.

1.4 Production Technology and Constraints

The production function embodies realistic constraints and decision-making processes:

Leontief Technology with Multiple Limiting Factors:** *A producer planning to manufacture a quantity Q of output must satisfy several simultaneous requirements. First, for each intermediate input type i , the producer must hold inventory of at least $Q \times n_i$ units, where n_i represents the technical coefficient from the SAM. Second, labor requirements must be met by hiring workers for $Q \times L$ hours, where L represents compensation of employees per unit output divided by the wage rate. Third, capital requirements dictate that the producer's fixed capital stock K must satisfy $K \geq Q \times k$, where k represents gross operating surplus per unit converted to capital units through a productivity factor.*

The actual producible quantity is determined by the minimum across all these constraints. If the producer lacks sufficient intermediate inputs, labor, or capital, production is reduced to the maximum feasible level given the tightest constraint. This generates realistic capacity utilization dynamics and creates demand for factor accumulation.

Adaptive Production Planning:** *Producers do not optimize with perfect foresight but instead rely on adaptive expectations. Each month, firms update a stock reference level representing their desired inventory, calculated as $S = D \times \phi$, where D is the moving average of recent demand and ϕ is a target stock coverage parameter. The production target for the next period is then set to $P = \max(0, S - I)$, where I represents current inventory. This rule generates realistic inventory cycles and implies that firms produce to replenish stocks rather than attempting to predict future sales with precision.*

Dynamic Capacity Adjustment:** *When a firm consistently exhausts its inventory, signaling sustained excess demand, it gradually increases its stock reference. Conversely, persistent unsold inventory leads to downward adjustment. Capital stock evolves through gross fixed capital formation, where firms convert finished goods into fixed capital using GFCF composition vectors from the SAM. Labor capacity adjusts through hiring and firing decisions made before each production cycle.*

1.5 Market Mechanisms and Price Discovery

Markets operate through decentralized bilateral matching rather than Walrasian auctioneers:

Goods Markets: Each simulation period is divided into workdays. On each workday, a randomized subset of agents engages in trade. Buyers search their neighborhood for sellers offering desired products. When a buyer encounters a seller, trade occurs if the buyer's willingness to pay equals or exceeds the seller's asking price. The buyer's willingness to pay adapts upward after failed searches and downward after successful purchases. The seller's asking price adapts upward after successful sales and downward toward production cost after failed sales. This double adaptation process gradually aligns prices with market clearing levels without requiring centralized coordination.

Labor Markets: Workers post wage demands based on their recent experience and inflation expectations. Producers post wage offers based on their profitability and hiring urgency. Employment relationships form when a worker's wage demand does not exceed a producer's offer. The matching process occurs through neighborhood search, creating geographic wage variation and realistic search unemployment. Workers who remain unemployed for extended periods gradually reduce wage demands but not below inflation-adjusted levels, generating a natural unemployment rate that emerges from the search process rather than being imposed exogenously.

Financial Markets: Primary equity markets allow firms to issue shares to finance growth, with initial shares typically held by the founding entrepreneur. Secondary markets enable workers to trade shares among themselves and with firms, though this feature is parameterized and may be disabled. Banks provide credit to firms based on assessed default risk, charging interest rates that incorporate a base rate, a bank markup, and a risk premium calculated from the firm's equity position. The credit market thus rations financing to firms based on their financial strength, creating realistic pecking order financing behavior.

1.6 Government and External Sectors

Government Operations: The government maintains a comprehensive fiscal account. On the revenue side, it collects production taxes levied on gross output by sector, product taxes applied to final consumption, and labor taxes derived from compensation of employees. On the expenditure side, it purchases goods and services from domestic producers according to SAM-specified consumption patterns, makes transfer payments when unemployment insurance is activated, and services public debt through interest payments to the central bank.

Government consumption is calibrated during the adjustment period through adaptive rules that scale consumption factors for each product type to achieve target expenditure levels. Tax rates are fixed at SAM-derived values but can be subjected to policy experiments during specified periods. Public debt management occurs automatically, with the government issuing bonds to the central bank to finance any deficits and accumulating reserves during surpluses.

External Sector Modeling: Each external sector represents either an individual foreign country (when detailed bilateral trade data is available) or an aggregate of trading partners. The external sector maintains export and import vectors tracking bilateral trade flows with the domestic economy. Imports are supplied elastically at exogenously determined prices, while export demand follows patterns derived from the target SAM but can evolve dynamically if inter-temporal input-output data is provided.

This approach allows the simulation of open economy dynamics including trade balance evolution, terms of trade effects, and the impact of external demand shocks. The calibration process adjusts trade propensities to match empirical trade patterns during the initialization phase.

PART II: OPERATIONAL DYNAMICS

2.1 Initialization and Bootstrap Sequence

The simulation begins with an economy devoid of production structure:

Phase 0 (Month -1): Pre-Initialization: The system loads the target SAM from input-output tables, typically sourced from FIGARO or national accounts databases. From this SAM, it extracts the number of production sectors, defines product specifications for each sector including technical coefficients and factor requirements, establishes the government consumption vector and tax structures, identifies external trade partners and bilateral trade patterns, and initializes global parameters including the number of workers to simulate as a scaled-down representation of the actual labor force.

Workers are instantiated with identifiers ranging from 0 to $N-1$, where N is the target worker population. Each worker is assigned a household group (if the SAM disaggregates households) and positioned in the neighborhood network. Workers begin with zero wealth, zero employment, and no goods inventory. They are endowed with one unit of available labor time and initial wage expectations set equal to the SAM-derived initial salary.

Phase 1 (Months 0 to T_{assist}): Assisted Production Period**: *This critical phase enables the economy to develop production capacity. Since no firms exist initially, unemployment would be 100% and production would be zero without intervention. The assisted production mechanism provides temporary support through two channels.*

First, whenever a production sector has zero active firms during this period, the system probabilistically creates a new firm in that sector with high probability, ensuring that at least one firm exists for each necessary production activity. These startup firms are assigned to wealthy workers who have accumulated sufficient resources, or created as government-owned enterprises for public sectors.

Second, and most importantly, if a firm produces quantity Q in a period but targeted production was $Q_{\text{target}} > Q$ due to insufficient inputs, labor, or capital, the system adds an artificial quantity $\Delta = (Q_{\text{target}} - Q) \times f(t)$ to the firm's output, where $f(t) = (T_{\text{assist}} - t) / T_{\text{assist}}$ is a linearly declining support factor. This assisted production appears in the firm's inventory and is distributed to buyers in proportion to their consumption patterns derived from the SAM.

The assistance mechanism prevents the chicken-and-egg problem where firms cannot produce because workers have no income to buy products, while workers have no income because firms cannot hire them. By temporarily subsidizing production, the system allows an initial accumulation of goods inventories, enables worker employment and income generation, facilitates gradual capital accumulation by firms, and permits price discovery processes to begin functioning.

Assisted production tapers linearly to zero over the support period, typically spanning 40-50 months (representing approximately 4 years of simulation time). The declining support forces firms to become progressively more self-sufficient, increasing competitive pressure over time.

2.2 Calibration Period: Guiding Toward Steady State

Following the assisted production phase, the system enters an extended calibration period designed to align emergent simulation dynamics with empirical targets:

Wealth Calibration (Months T_wealth_start to T_wealth_end):** Worker wealth is continuously monitored and adjusted. The target wealth level W_target is derived from the SAM's household financial assets and gross fixed capital formation. Each period, the system calculates current aggregate worker wealth $W_current$ through direct summation of cash holdings, financial asset values, and residential capital stocks. If $W_current$ deviates from W_target , the system applies a correction. When wealth is excessive, worker cash is reduced uniformly; when deficient, workers receive proportional cash injections. This process, implemented in `CWorld::AdjustSAMParameters()`, typically occurs over 50-60 months.

The wealth adjustment serves multiple purposes. It ensures that consumption demand remains consistent with SAM-implied levels, prevents unrealistic asset accumulation during the transition, maintains proper capital-output ratios as production scales up, and anchors financial wealth to empirically observed levels.

Consumption Calibration (Ongoing During $T_calibration$):** Household and government consumption propensities require fine-tuning to match SAM expenditure patterns. For each product type g , the system maintains a consumption factor ϕ_g that scales an agent's consumption decision. During calibration, these factors adjust adaptively. For households, the rule is $\phi_g(t+1) = (1-\alpha)\phi_g(t) + \alpha \times \phi_g(t) \times [1 - \varepsilon_g(t)]$, where $\varepsilon_g(t) = [C_g(t) - C_g^{target}] / C_g^{target}$ represents the consumption error and α is an adjustment speed parameter.

This adaptive rule gradually brings consumption closer to targets without generating oscillations. Similar adjustments occur for government consumption. The calibration accounts for the fact that workers' consumption decisions depend on wealth levels, while the system simultaneously adjusts wealth stocks. The process therefore requires careful balancing to prevent circular dependencies.

Unemployment Targeting (Months T_unemp_start to $T_calibration$):** A crucial aspect of calibration involves achieving the target unemployment rate specified in the SAM, typically ranging from 4% to 8% for developed economies. The system adjusts the `InitSalaryCalibFactor`, a global multiplier applied to wage levels. The adjustment rule is $CalibFactor(t+1) = CalibFactor(t) \times [1 + \beta \times (U_target - U_current)]$, where U_target is the SAM's initial unemployment rate, $U_current$ is the simulated unemployment rate, and β is a sensitivity parameter.

When simulated unemployment exceeds the target, wages are scaled downward to incentivize hiring. When unemployment falls below target, wages rise to restrain labor demand. This mechanism operates analogously to monetary policy rules in actual economies, where central banks adjust interest rates to achieve unemployment targets. The calibration factor typically stabilizes near unity after 50-60 months, indicating that emergent wage-setting behavior aligns with empirical compensation levels.

Price Liberalization and Inflation Introduction:** During early calibration months, product prices and wages may be held fixed or adjusted only gradually to maintain stability. Once basic calibration has progressed (typically after 60-120 months), price adjustment mechanisms are fully activated. Firms begin to adjust markups in response to excess demand or supply. If a firm's inventory falls below a threshold fraction of its reference stock, and demand is growing, the firm increases its markup factor by an inflation parameter (typically 1.002, implying 0.2% price increase). This generates endogenous inflation that mimics cost-push and demand-pull dynamics observed in real economies.

The inflation mechanism serves dual purposes. It tests the economy's resilience to monetary disturbances and creates realistic relative price adjustments that facilitate structural change and factor reallocation.

2.3 Post-Calibration Steady-State Dynamics

After calibration concludes (typically month 600-720, representing 50-60 years), the system operates with minimal interventions:

Endogenous Firm Demography**: *New firms continue to enter when workers accumulate sufficient wealth (typically 8-12 times monthly salary) and perceive opportunities. Entry probability depends on local neighborhood demand conditions. Entrepreneurs examine recent consumption patterns in their vicinity and preferentially start firms in sectors with high unmet demand. This generates realistic spatial clustering and sectoral entry patterns.*

Firm exit occurs through insolvency. When a firm's wealth becomes negative and remains negative for a specified number of consecutive periods (typically 3-6 months), the firm is dismantled. Its remaining assets are liquidated, employees are released to the labor pool, and the firm's identifier becomes available for new entrants. The calibration parameters ensure that entry and exit rates stabilize near empirically observed levels (typically 5-10% annually), indicating a healthy Darwinian churn.

Growth Dynamics: Once calibrated, the economy exhibits quasi-steady-state growth. GDP increases primarily through two channels: productivity improvements that emerge endogenously as successful firm types expand relative to less efficient competitors, and capital deepening as gross fixed capital formation accumulates productive capacity. The system does not impose exogenous technological progress functions. Instead, productivity growth emerges from the Darwinian selection mechanism where more efficient firms grow while less efficient firms shrink or exit.

The growth rate stabilizes around empirically realistic levels (typically 2-3% annually for developed economies) without explicit calibration of growth targets. This emergent growth represents a key validation of the model's ability to generate realistic long-run dynamics from microeconomic behavioral rules.

Business Cycle Fluctuations**: *Even in steady state, the economy exhibits cyclical fluctuations. Inventory dynamics generate short-term oscillations. When many firms simultaneously overproduce in response to high demand, aggregate inventory accumulation can temporarily depress production in subsequent periods. Conversely, synchronized inventory depletion stimulates production surges. These cycles typically have periodicities of 6-18 months, resembling empirical inventory cycles.*

Credit cycles also emerge. When banks simultaneously tighten lending in response to perceived risk, firm liquidity can become constrained, depressing investment and production. The subsequent reduction in default risk encourages renewed lending, initiating expansion. These financial cycles interact with real cycles to generate rich fluctuation dynamics without requiring exogenous demand or supply shocks.

2.4 Darwinian Selection Mechanisms in Operation

The Darwinian aspect manifests through several interlocking processes:

Differential Replication: Successful firms expand by accumulating capital, hiring additional workers, and increasing production scale. As they grow, they become more visible in the neighborhood network, attracting more customers and workers. Their success enables them to pay competitive wages and maintain sufficient

inventories, further reinforcing their competitive position. This creates a positive feedback loop where fitness (defined by profitability and growth) begets further fitness.

Variation Generation: Heterogeneity arises from multiple sources. Firms are created at different times and thus face different competitive environments. Random assignment of location in the neighborhood network creates geographic variation in local demand conditions. Stochastic shocks to demand, supply, and matching processes ensure that even identical firms in similar locations experience different trajectories. Workers vary in their wealth accumulation rates, risk tolerance, and entrepreneurial timing, generating diversity in firm founding patterns.

Selection Pressure: The insolvency mechanism provides the selection force. Firms must maintain positive cumulative wealth over time to survive. Those with inefficient production, poor pricing strategies, or disadvantageous locations gradually lose wealth and eventually exit. The selection intensity is calibrated through parameters controlling the minimum number of firms per sector and the required duration of negative wealth before dismantling. Stronger selection pressures accelerate structural change but may destabilize the system; weaker pressures allow marginal firms to persist, reducing turnover.

Heritability and Learning: Although the system does not implement explicit evolutionary algorithms with mutation and crossover, a form of heritability exists through imitation and learning. Successful pricing strategies, production scales, and hiring policies implicitly diffuse through the economy as new entrants observe market conditions and set parameters accordingly. The adaptive price and wage adjustment rules embody a form of reinforcement learning where behaviors that generate profits are amplified.

2.5 Key Control Parameters and Sensitivity

Several parameters critically influence system behavior:

AssistedProductionUpto: Determines the duration of production assistance. Shorter periods (20-30 months) accelerate the transition to market-determined production but risk systemic instability if the economy has not developed sufficient productive capacity. Longer periods (60-80 months) provide more gradual transition but may delay the emergence of genuine market competition.

FinishCalibrationAt: Sets the endpoint of active parameter adjustment. Extending calibration beyond 600 months generally provides marginal improvement while increasing computational burden. Terminating calibration prematurely (before 400 months) often results in persistent deviations from SAM targets.

ProductionChangeFraction: Controls the speed at which firms adjust production plans in response to demand signals. Higher values (0.4-0.6) make firms more responsive but increase volatility. Lower values (0.1-0.3) smooth fluctuations but slow convergence.

ConsumAdjustFraction: Governs the adaptation rate of consumption factors during calibration. This parameter must be carefully tuned; excessive adjustment speeds cause oscillations, while insufficient adjustment prolongs convergence.

MonthsWithNegativeWealthToDismantle:** *Determines selection intensity. Requiring 3-4 consecutive months of negative wealth provides rapid turnover and strong selection. Requiring 8-12 months allows struggling firms more time to recover, reducing volatility but potentially perpetuating inefficiency.*

*MinSalariesToStartProducer***: Controls the wealth threshold for entrepreneurship. Lower thresholds ($3-5 \times \text{salary}$) facilitate entry and increase firm numbers, while higher thresholds ($10-15 \times \text{salary}$) concentrate entrepreneurship among wealthier workers and reduce firm counts.

2.6 Monthly Operational Cycle

Each simulation month proceeds through a structured sequence:

Day 0: Month Initialization: Statistics are reset for the new period. Firm demographics are updated through entry and exit processes. The `DismantleAndStartNewProducers()` function identifies firms with consecutive negative wealth exceeding the threshold and removes them. Workers with sufficient wealth and entrepreneurial propensity create new firms in sectors with perceived demand. The SAM adjustment function applies wealth, consumption, and wage calibration updates if within the calibration window. Monthly data structures are cleared and reinitialized.

Days 1-20: Production and Trading Workdays: The month is divided into 20 workdays. Each workday follows an internally structured sequence. First, agents are shuffled into random order to prevent systematic biases from processing sequence. Second, selected agents engage in production activities if this workday is designated as a production day according to their production schedule. Third, procurement activities occur as firms search for intermediate inputs in their neighborhoods. Fourth, consumption activities proceed as workers search for final goods and services. Fifth, labor market interactions happen as unemployed workers search for jobs and firms with vacancies search for workers. Sixth, financial market activities including share trading and loan applications are processed. This daily iteration creates realistic short-term dynamics and allows information to propagate gradually through the economy.

End of Month: Aggregation and Adjustment: After all workdays conclude, comprehensive monthly statistics are compiled. The `AnalyzeLastMonth()` function in `CData` aggregates transactions into monthly SAM matrices, calculates GDP from expenditure and income approaches, computes inflation rates from price indices, measures unemployment and labor market statistics, records firm demographics and size distributions, evaluates wealth distributions across households and firms, and assesses financial sector indicators. These statistics inform both the calibration adjustments for the next period and the output data for analysis.

2.7 Convergence Characteristics and Validation

The system typically requires 600-900 simulation months to reach quasi-steady state, representing 50-75 years of simulated time. Convergence is evaluated through multiple criteria:

SAM Distance Metric: Monthly transactions are aggregated into a simulated SAM matrix and compared with the target SAM. The Euclidean distance between matrices, normalized by the Frobenius norm of the target, serves as a primary convergence indicator. Successful calibration typically achieves distance metrics below 5-10%, with most cells matching within 15% of target values.

Macroeconomic Stability: Key aggregate variables including GDP, unemployment, inflation, and trade balance should stabilize around steady-state means with fluctuations of realistic magnitude. Typical volatility after calibration shows GDP growth rates varying $\pm 1-2$ percentage points around mean growth, unemployment fluctuating $\pm 0.5-1.5$ percentage points around the natural rate, and inflation remaining bounded within $\pm 0.5\%$ annually.

Microeconomic Realism: Firm size distributions should approximately follow Zipf's law (power law distributions) consistent with empirical evidence. Entry and exit rates should match economy-wide averages from business demography statistics. Wage and profit rate distributions should exhibit realistic dispersion with most firms earning positive but modest returns and a tail of highly profitable enterprises.

Robustness Tests: The system's validity is assessed through perturbation experiments. After achieving steady state, exogenous shocks such as demand reductions in specific sectors, tax policy changes, or trade disruptions are introduced. The economy's response including adjustment speed, sectoral reallocation patterns, and return to equilibrium provides validation that the simulation captures realistic adjustment mechanisms.

PART III: THEORETICAL AND METHODOLOGICAL IMPLICATIONS

3.1 Departure from Standard Macroeconomic Frameworks

This simulator fundamentally departs from established macroeconomic modeling approaches:

DSGE Models: Dynamic Stochastic General Equilibrium models assume representative agents with rational expectations, market-clearing conditions enforced continuously, and aggregate behavior derived analytically from microfoundations. DEPLOYERS instead implements heterogeneous agents with adaptive expectations, persistent market disequilibria and gradual adjustment, and aggregate behavior that emerges computationally rather than being assumed a priori. While DSGE models trace shocks through analytic impulse-response functions, DEPLOYERS generates responses through agent interactions.

Input-Output Models: Traditional Leontief input-output analysis assumes fixed technical coefficients, instantaneous adjustment to demand changes, and no behavioral dynamics. DEPLOYERS uses I-O tables as initial conditions but allows technical coefficients to evolve, implements gradual adjustment through inventory and capital dynamics, and generates rich behavioral dynamics including firm entry, exit, and growth. I-O models provide static multipliers; DEPLOYERS simulates dynamic adjustment paths.

Computational General Equilibrium:** *CGE models retain the equilibrium assumption while allowing for more detailed sectoral disaggregation. They typically solve for equilibrium prices and quantities that clear all markets simultaneously. DEPLOYERS abandons equilibrium as an organizing principle, instead using iterative bilateral trading that generates equilibration tendencies without assuming equilibrium is ever reached. CGE models are comparative statics devices; DEPLOYERS is fundamentally dynamic.*

3.2 Common-Sense Agents vs. Theoretical Optimization

The behavioral rules embody intuitive heuristics rather than utility maximization:

Firms: Real firms typically use markup pricing (cost plus margin) rather than equating marginal cost to marginal revenue. They maintain target inventory-to-sales ratios rather than optimizing stochastic inventory models. They adjust prices gradually in response to persistent excess demand or supply rather than continuously market-clearing. They hire until their order book requires it rather than solving dynamic programming problems for optimal labor demand. DEPLOYERS captures these realistic heuristics.

Workers: Actual households often save a roughly constant fraction of income rather than solving life-cycle optimization problems with Euler equations. They adjust consumption based on recent income experience rather than forming expectations over infinite horizons. They search for jobs locally rather than conducting global job searches. They often remain with employers that provide satisfactory compensation rather than continuously seeking the highest wage. These patterns appear in the simulation.

System-Level Implications: By using heuristic rules that real agents might plausibly employ, the simulator achieves a form of "ecological rationality" where simple rules prove robust across diverse environments. This contrasts with optimization-based models where agents solve unrealistically complex problems. The emergent macroeconomic patterns thus reflect genuine microeconomic behavior rather than artifacts of convenient optimizing assumptions.

3.3 Evolutionary Economics Perspective

The Darwinian framework connects to evolutionary economics traditions:

Nelson-Winter Evolutionary Theory:** *Richard Nelson and Sidney Winter pioneered evolutionary models where firms use routines (repeated behavior patterns) rather than optimizing, compete for market share based on relative efficiency, and face selection pressures that eliminate unsuccessful routines. DEPLOYERS implements these concepts with firms employing simple production and pricing routines, competing through price and availability rather than strategic optimization, and exiting when routines prove unprofitable. The system generates creative destruction dynamics where new firms replace old without requiring exogenous technological shocks.*

Schumpeterian Growth: Schumpeter emphasized entrepreneurship, innovation, and creative destruction as growth drivers. DEPLOYERS generates entrepreneurial entry based on perceived opportunities, structural change through entry and exit rather than representative agent reallocation, and productivity growth from selection among heterogeneous producers. While the current implementation does not include explicit innovation, the framework could naturally accommodate Schumpeterian innovation by allowing successful firms to occasionally improve their production coefficients.

Complex Adaptive Systems: The simulation embodies characteristics of complex adaptive systems: many interacting heterogeneous agents, nonlinear feedbacks generating emergent properties, no central coordinator or equilibrium assumption, and robust yet evolving structures. The economy exhibits typical CAS features including power law distributions in firm sizes, punctuated equilibrium where long periods of stability are interrupted by structural shifts, and resilience to perturbations through distributed adaptability.

3.4 Calibration as an Empirical Method

The calibration approach deserves methodological attention:

Indirect Inference: By adjusting parameters until simulated statistics match empirical targets, the calibration procedure resembles indirect inference, a statistical method where model parameters are estimated by matching simulated and actual moments. The SAM provides a comprehensive set of moments including sectoral outputs, inter-industry flows, factor payments, and final demands. The calibration thus fits the model to a rich multidimensional empirical target.

Structural Identification: Because the SAM over-identifies the model (many more empirical moments than behavioral parameters), the system faces genuine identification challenges. Multiple parameter combinations might generate similar aggregate outcomes. The calibration procedure effectively searches this parameter space for regions that produce stable, realistic behavior. This parallels structural econometric identification where multiple structural models may be observationally equivalent.

Validation Strategy: A critical test of the methodology involves out-of-sample validation. After calibrating to a specific year's SAM, the model's predictions for subsequent years (when available from longitudinal I-O tables) can be compared with actual evolution. If the model reasonably tracks structural change without recalibration, this validates both the behavioral rules and the calibration strategy. Such validation remains an important direction for future research.

3.5 Computational Efficiency and Scalability

The system's computational characteristics merit consideration:

Complexity Scaling: With N workers and M producer types, computational cost scales approximately as $O(N \times M)$ per simulation month since each agent potentially interacts with multiple partners across product markets. For typical simulations with $N \approx 10,000$ workers and $M \approx 20$ sectors, processing speeds achieve 100-500 months per hour on modern hardware. The C++ implementation provides substantial performance advantages over interpreted languages like Python.

Parallelization Potential: Many operations within a simulation day can be parallelized. Bilateral trades between different pairs of agents are independent and could occur concurrently. Production activities across different firms are similarly independent. The current implementation uses primarily sequential processing but could be refactored for parallel execution, potentially achieving order-of-magnitude speedups on multi-core processors or GPU architectures.

Memory Efficiency: The system maintains relatively modest memory footprints despite detailed micro-level state tracking. By storing only essential variables and using efficient data structures (sparse matrices for SAM representation, hash maps for agent lookups), typical simulations consume 1-5 GB of RAM. This enables desktop execution without requiring high-performance computing resources.

PART IV: INSIGHTS FROM THE DARWINIAN MECHANISM

4.1 The Role of Assisted Production

The assisted production phase proves essential for system bootstrapping:

Preventing Collapse: Without assistance, the economy would face a coordination failure. Firms cannot produce without inputs, but input suppliers cannot operate without demand for their outputs. Workers cannot consume without income, but firms cannot pay wages without revenue. This circular dependency would prevent any economic activity from emerging. Assisted production breaks this vicious cycle by temporarily providing exogenous support.

Gradual Withdrawal: The linear tapering of assistance forces firms to progressively increase efficiency. In early months, a firm might receive 80-90% of its output through assistance with only 10-20% from actual production. By the end of the assistance period, the firm must produce 100% through its own efforts. This graduated transition prevents sudden shocks that could destabilize the system.

Differential Impact: Not all firms receive equal assistance. Those producing high-demand consumer goods stabilize quickly as households begin earning income and purchasing products. Firms in upstream sectors producing intermediate inputs face longer development periods since demand for their products depends on downstream producers' growth. The assistance mechanism thus implicitly provides more support to struggling upstream sectors.

4.2 Emergence of Market Structure

The Darwinian process generates realistic industry structures:

Concentrated vs. Fragmented Sectors:** *Sectors characterized by high capital requirements and strong economies of scale tend to develop concentrated structures with few large firms. Conversely, sectors with low entry barriers and minimal scale economies become fragmented with many small producers. These patterns emerge endogenously from the interaction of production technology, entry costs, and selection pressures, matching empirical industry structures without requiring exogenous assumptions about market forms.*

Geographic Clustering: The neighborhood network structure combined with local demand creation generates spatial clustering. Successful firms attract workers to their vicinity, creating local demand for consumer goods and thus opportunities for additional firms. This positive agglomeration feedback mirrors real urban economics, where employment centers spawn supporting services and residential development.

Turnover Dynamics: Even after reaching steady state, constant turnover occurs with firm entry and exit. This churn proves essential for long-run efficiency. New entrants experiment with different scales, locations, and strategies. Successful experiments proliferate; unsuccessful ones quickly exit. This ongoing experimentation and selection mechanism generates continuous improvements in aggregate efficiency.

4.3 Labor Market Dynamics

The labor market exhibits rich dynamics:

Search Unemployment: Even when aggregate labor demand equals supply, unemployment persists due to search frictions. Workers need time to locate suitable employers, negotiate wages, and transition between jobs. The neighborhood search structure creates geographic mismatch where vacancies exist in one location while unemployed workers reside elsewhere. This generates realistic Beveridge curves (unemployment vs. vacancy relationships) without requiring rigid wages or exogenous matching functions.

Wage Dispersion: Identical workers employed in different firms receive different wages due to firm heterogeneity. More productive firms pay higher wages while marginal firms pay less. Workers with longer job tenure often receive higher compensation through accumulated informal negotiations. This dispersion matches empirical wage distributions and arises naturally from decentralized bargaining rather than being imposed through exogenous wage schedules.

Employment Dynamics: The system generates realistic hiring and separation flows. Monthly job creation and destruction rates fluctuate around mean levels of 2-4%, consistent with empirical labor flow data. During expansions, hiring accelerates and separations decline; during contractions, the pattern reverses. These procyclical employment flows emerge from the interaction of production dynamics and labor demand without requiring explicit business cycle mechanisms.

4.4 Financial Dynamics and Crises

Financial sector behavior generates additional insights:

Credit Rationing: Banks ration credit based on assessed default risk. Firms with negative equity or weak balance sheets face reduced credit access or higher interest rates. This creates realistic pecking order financing where firms prefer internal funds, then debt, and finally equity. The credit rationing mechanism also generates financial accelerator effects where small shocks become amplified through tightening credit conditions.

Systemic Risk Emergence: Correlated firm failures can generate systemic instability. If many firms simultaneously experience negative demand shocks, widespread losses erode bank capital, tightening credit conditions economy-wide. This creates contagion dynamics where initial losses propagate through the financial system. While the current implementation maintains conservative bank capital requirements to prevent systemic crises, the architecture could support research on financial stability policies.

Interest Rate Effects: The central bank's base rate influences commercial lending rates and thus firm investment decisions. Higher base rates increase financing costs, reducing capital accumulation and growth. Lower rates stimulate investment but may encourage excessive leverage. These transmission mechanisms operate through realistic channels of credit demand and supply rather than through assumed Euler equation responses.

4.5 Calibration Period as Policy Transition

The calibration process can be interpreted as a policy intervention:

Temporary Stabilization: The wealth adjustments and consumption scaling during calibration resemble temporary fiscal transfers or monetary accommodation that stabilizes the economy during structural transformation. Just as actual economies undergoing transition may receive IMF support or implement stimulus programs, the simulation employs calibration adjustments to facilitate orderly development.

Institutional Development: In reality, functioning market economies require institutional infrastructure including property rights, contract enforcement, financial regulation, and social safety nets. The calibration period implicitly represents the time required for these institutions to develop. Early in calibration, strong interventions stabilize the system; later, interventions diminish as organic market mechanisms strengthen.

Path Dependence: Initial conditions and calibration sequences affect long-run outcomes. Different random seeds generate different firm entry sequences and competitive outcomes, creating path-dependent trajectories. This mirrors historical economic development where initial conditions, institutional choices, and random events shape long-run structures. The simulation thus captures essential features of historical path dependence.

PART V: APPLICATIONS AND FUTURE EXTENSIONS

5.1 Policy Simulation Experiments

The framework enables diverse policy analyses:

Fiscal Policy: Tax experiments can examine how changes in production taxes, product taxes, or labor taxes affect employment, output, and welfare. The system captures both direct effects on government revenue and indirect general equilibrium effects through changes in prices, production, and consumption. Unlike static CGE models, DEPLOYERS traces dynamic adjustment paths showing how long policy changes take to achieve full effects.

Trade Policy: Tariff changes or trade agreement simulations can be implemented by adjusting external sector parameters. The model can examine how import restrictions affect domestic production, employment in import-competing sectors, consumer prices, and retaliation effects on exports. The neighborhood structure allows analysis of geographic incidence where some regions benefit while others suffer from trade policies.

Labor Market Regulation: Minimum wage experiments, unemployment insurance programs, and hiring subsidies can be tested. The decentralized labor market structure allows realistic evaluation of how regulations affect employment, wages, and worker welfare. For instance, minimum wages that bind only for marginal firms might reduce employment less than general equilibrium models suggest by accelerating creative destruction that shifts workers to more productive employers.

Industrial Policy: Sector-specific interventions including production subsidies, preferential credit access, or investment tax credits can be evaluated. The evolutionary framework permits examination of whether temporary assistance allows infant industries to reach competitive scale or whether protection merely perpetuates inefficiency. The long-run effect depends on whether protected firms become genuinely productive or remain dependent on support.

5.2 Extensions to the Current Framework

Several enhancements could enrich the model:

Technological Change: Implementing endogenous productivity improvements would complete the Schumpeterian growth mechanism. Successful firms could occasionally improve their technical coefficients through R&D; investment. The probability of innovation could depend on firm size, profitability, and sector characteristics. This would generate realistic productivity growth patterns and allow analysis of innovation policies.

Human Capital: Workers could accumulate skills through education and experience, affecting their productivity and wages. Firms could invest in worker training. This would create realistic wage-experience profiles, skill mismatch unemployment, and human capital accumulation dynamics. Educational policy experiments including subsidy programs or loan availability could be examined.

Environmental Constraints: Adding pollution emissions linked to production and consumption, along with emissions taxes or cap-and-trade systems, would enable climate policy analysis. The microeconomic adjustment mechanisms would show how sectors substitute away from dirty inputs, how firms invest in clean technology, and how carbon pricing affects employment and competitiveness.

Financial Innovation: Expanding the financial sector to include corporate bonds, mortgage markets, and securitization would enable analysis of financial development and regulation. The 2008 financial crisis demonstrates the importance of financial market structure for real economy outcomes. A richer financial sector would allow examination of financial stability policies and macroprudential regulation.

Multiple Regions: Implementing multiple domestic regions with internal trade would enable spatial policy analysis. Regional development policies, transport infrastructure investments, and agglomeration economies could be studied. This would connect to economic geography literature examining why some regions prosper while others stagnate.

5.3 Validation and Empirical Testing

Rigorous empirical validation remains crucial:

Moment Matching: Beyond matching the SAM, the model should generate realistic business cycle statistics including output volatility, consumption smoothness, and investment procyclicality. Frequency domain analysis could compare simulated and actual power spectra of macro variables. Matching second moments (volatilities and correlations) as well as first moments (means) would strengthen validation.

Micro-Data Comparison: Comparing simulated firm-level and worker-level distributions with administrative data provides additional validation. Firm size distributions, growth rate distributions, survival rates by age and size, wage distributions, job tenure distributions, and wealth distributions should match empirical patterns. Deviations would indicate where behavioral rules require refinement.

Out-of-Sample Prediction: The strongest validation would involve calibrating to one year's SAM and testing whether the model predicts subsequent years' structural evolution. With longitudinal I-O tables available for many countries, this validation is feasible. Successful prediction would demonstrate that the behavioral rules capture genuine economic mechanisms rather than merely fitting cross-sectional data.

Natural Experiments: Historical natural experiments including policy changes, trade liberalizations, or technology shocks provide additional testing grounds. Calibrating the model to pre-intervention conditions and simulating the intervention should generate predictions comparable to econometric evaluations of actual interventions. This would test whether the model captures causal mechanisms.

5.4 Comparison with Alternative Modeling Approaches

The framework's advantages and limitations relative to alternatives deserve consideration:

Versus DSGE: DEPLOYERS captures richer heterogeneity and disequilibrium dynamics but sacrifices theoretical tractability. DSGE models provide clean welfare analysis through utility foundations; DEPLOYERS requires alternative welfare metrics. DSGE impulse responses are analytically derived; DEPLOYERS responses must be simulated. The choice depends on whether the research question requires micro-level detail

and disequilibrium dynamics or prefers analytical tractability.

Versus Stock-Flow Consistent Models**: *SFC models emphasize accounting consistency and financial-real interactions. DEPLOYERS shares the accounting rigor but adds microeconomic heterogeneity and behavioral dynamics. SFC models typically assume behavioral equations for sectoral aggregates; DEPLOYERS derives aggregate behavior from individual decisions. SFC models excel at tracking financial imbalances; DEPLOYERS excels at structural change and distributional dynamics.*

Versus Microsimulation: Tax-benefit microsimulation models apply policy rules to representative samples of actual households but typically lack behavioral response or general equilibrium effects. DEPLOYERS provides endogenous behavioral response and market clearing but uses stylized agents rather than representative samples. Hybrid approaches that combine microsimulation detail with ABM dynamics represent a promising direction.

5.5 Educational and Pedagogical Applications

Beyond research, the system offers educational value:

Teaching Economic Principles**: *Students can observe how markets coordinate activity, how prices emerge from decentralized interactions, how specialization and trade generate gains, and how institutions affect outcomes. Visualizing these dynamics in real time deepens understanding beyond textbook equilibrium diagrams.*

Experimental Economics: The simulation creates a laboratory for economic experiments without human subjects. Students can propose policy interventions, simulate outcomes, and interpret results. This active learning approach engages students and develops economic intuition. Competitive experiments between student teams designing optimal policies could motivate learning.

Complexity Economics: The model exemplifies complexity science principles including emergence, self-organization, adaptation, and power laws. Students interested in complex systems can see these concepts illustrated in an economically meaningful context. This interdisciplinary connection attracts students from physics, computer science, and engineering into economics.

PART VI: CONCLUSION

6.1 Summary of Key Contributions

DEPLOYERS advances macroeconomic modeling through several innovations:

Darwinian Foundation: By treating firm creation, competition, and survival as evolutionary processes, the model generates realistic industry dynamics without assuming equilibrium or imposing market structures. The economy develops endogenously from primitive conditions to complex production networks through organic processes.

Assisted Production Mechanism:** *The graduated support system solves the bootstrapping problem that prevents economy-wide coordination from emerging spontaneously. This insight may inform economic development policy, suggesting that temporary assistance allowing nascent industries to reach minimum viable scale can prove decisive.*

SAM-Based Calibration: Using comprehensive empirical targets from input-output tables grounds the simulation in real economic structures. The calibration methodology provides a template for developing realistic agent-based models that combine microeconomic behavioral detail with macroeconomic empirical discipline.

Common-Sense Agents: Implementing simple, realistic decision rules rather than optimization-based behavior makes the model more transparent and arguably more realistic. The emergence of sensible aggregate outcomes from simple micro-level rules demonstrates that rationality assumptions can be substantially relaxed without sacrificing explanatory power.

6.2 Theoretical Implications

The research suggests several theoretical insights:

Equilibrium as Emergent Property:** *Rather than assuming markets clear, the model shows equilibration tendencies emerging from adaptive behavior. Markets may approach but never precisely reach equilibrium. This perspective, more consistent with actual market observations, suggests economic theory should emphasize adjustment mechanisms rather than equilibrium states.*

Microfoundations Redux: The model provides microfoundations fundamentally different from those in DSGE models. Rather than deriving macroeconomic relationships from individual optimization, it derives them from simple heuristics and interaction protocols. This suggests an alternative approach to microfoundations based on behavioral realism rather than axiomatic rationality.

Growth as Selection: Productivity growth emerges from selection among heterogeneous producers rather than from exogenous technological progress. This evolutionary perspective connects to growth accounting evidence that reallocation across firms contributes substantially to aggregate productivity growth. Theory should perhaps emphasize selection and reallocation more and representative agent optimization less.

Coordination Without Coordinators:** *The system achieves remarkable coordination across thousands of interacting agents without any centralized authority or market-clearing mechanism. This*

demonstrates that decentralized economies can function through purely local interactions combined with adaptive learning. It suggests that institutional structures enabling local coordination may be more important than structures enforcing global market clearing.

6.3 Methodological Reflections

The methodology raises important questions about model validation and purpose:

Validation Standards: What constitutes sufficient validation for complex agent-based models? Matching moments proves necessary but perhaps insufficient. Predictive accuracy tests appear attractive but may be too demanding given the sensitivity of complex systems to initial conditions. Perhaps the appropriate standard is qualitative pattern matching—does the model generate the kinds of dynamics, distributions, and relationships observed in real economies?

***Explanatory vs. Predictive Goals**:** Lucas's famous critique argues that models with behavioral rules that shift with policy cannot provide policy guidance. Yet DEPLOYERS' behavioral rules remain fixed across policy regimes, with only the policy environment changing. This suggests that agent-based models with stable micro-level rules might escape Lucas critique problems that plague reduced-form models while avoiding the heroic rationality assumptions Lucas advocated.*

***Complementarity of Approaches**:** Perhaps different modeling approaches serve different purposes. DSGE models may excel at examining small perturbations around steady states where linearization proves accurate. Agent-based models may excel at examining large structural changes, distributional dynamics, and transition paths where disequilibrium matters. Rather than competitive alternatives, these approaches might be complementary tools addressing different questions.*

6.4 Future Research Directions

Several promising research directions emerge:

***Empirical Validation Studies**:** Systematic comparison of simulated and actual firm dynamics, worker transitions, and wage distributions using administrative data would strengthen the empirical foundations. Collaboration with statistical agencies to access confidential microdata for validation would prove valuable.*

Policy Applications: Developing specific policy simulation capabilities for tax reform, labor market interventions, and trade policies would demonstrate practical utility. Collaboration with policy institutions to inform actual policy debates would both test the model and contribute to evidence-based policymaking.

Theoretical Extensions: Incorporating innovation, human capital, and environmental constraints would enrich the framework. Developing general results about when evolutionary systems converge to efficient outcomes would contribute to evolutionary economic theory.

Methodological Refinement: Improving calibration algorithms, developing sensitivity analysis techniques adapted to complex nonlinear models, and creating visualization tools for understanding emergent dynamics would enhance the methodology's applicability.

6.5 Final Reflection

DEPLOYERS demonstrates that macroeconomic modeling can proceed from fundamentally different foundations than those dominating contemporary research. Rather than assuming rational expectations, market clearing, and representative agents, the model builds from heterogeneous agents with adaptive expectations, disequilibrium with gradual adjustment, and Darwinian selection processes. The result is a system that captures important aspects of economic reality while remaining computationally tractable and empirically grounded.

The research suggests that economics might benefit from embracing methodological pluralism. Different modeling approaches illuminate different aspects of economic phenomena. DSGE models provide analytical tractability and clean welfare analysis. Structural econometric models enable causal inference from observational data. Agent-based models capture heterogeneity, disequilibrium, and complex dynamics. Rather than seeking a single modeling paradigm, the discipline might progress faster by developing multiple complementary approaches, each suited to particular questions.

The Darwinian perspective also carries philosophical implications. It suggests that economies are better understood as evolutionary systems than as equilibrium systems. This view emphasizes process over state, adaptation over optimization, and emergence over design. Such a perspective may prove particularly relevant for understanding economic development, structural transformation, and institutional change—phenomena where evolutionary dynamics appear central.

Ultimately, DEPLOYERS contributes to a growing body of research demonstrating that agent-based computational models can serve as serious tools for macroeconomic analysis. These models complement rather than replace traditional analytical approaches. As computational power continues advancing and methods continue improving, agent-based models will likely play increasingly important roles in policy analysis and economic understanding. The Darwinian approach presented here represents one pathway forward in this broader methodological evolution.

Document prepared: November 2025 **Analysis of:** DEPLOYERS v2 - Agent-Based Macroeconomic Simulator
Core modules analyzed: SAM.cpp, World.cpp, Producer.cpp, Individual.cpp, CData.cpp, Agent.cpp, Banks.cpp, Government.cpp, and related components